

Automated SSH Brute Force Detection using Splunk SIEM and Shuffle SOAR

Project Overview

This project demonstrates a complete Security Operations Center (SOC) use case where an SSH brute-force attack is detected in real time using Splunk SIEM and automatically processed through Shuffle SOAR.

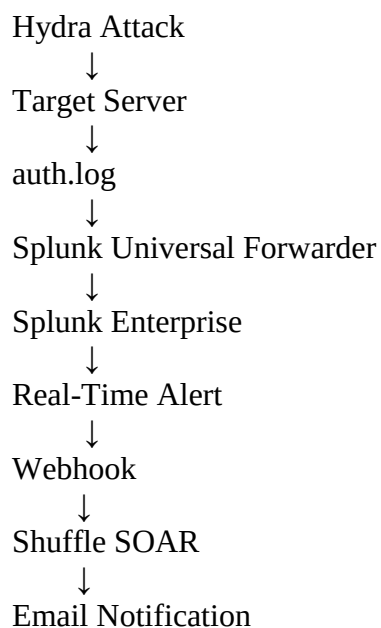
The goal was to simulate a real-world attack scenario, collect authentication logs, detect malicious activity, trigger alerts, and automate incident notification without manual intervention.

Project Architecture

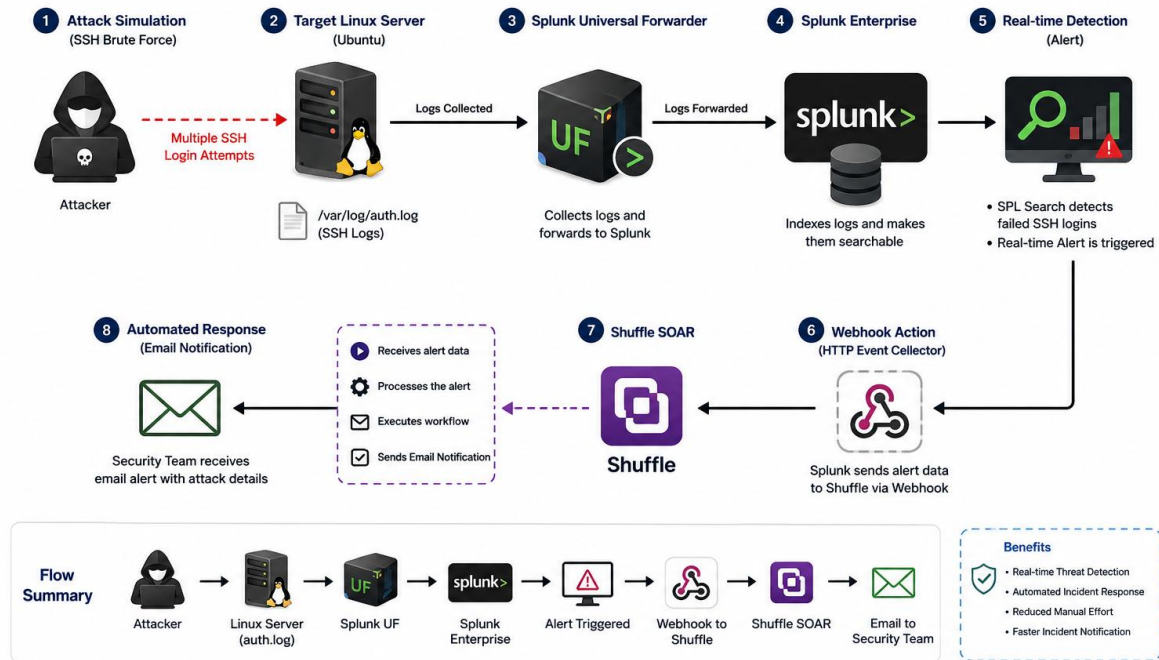
The environment consists of four machines:

Machine	IP Address	Purpose
Splunk SIEM	192.168.0.5	Log Collection & Detection
Target Server	192.168.0.6	Victim System
Attacker Server	192.168.0.7	Hydra Attack Machine
Shuffle SOAR	192.168.0.8	Automated Response

Workflow



Architecture – Automated SSH Brute Force Detection & Response using Splunk & Shuffle SOAR



Phase 1 - Splunk SIEM Setup

Splunk Server Deployment

A dedicated Ubuntu server was deployed to host Splunk Enterprise.

Configuration

VM Name: splunk-server

IP Address: 192.168.0.5

Operating System: Ubuntu Server

Purpose: Centralized Security Monitoring

Splunk Enterprise was installed and configured successfully.

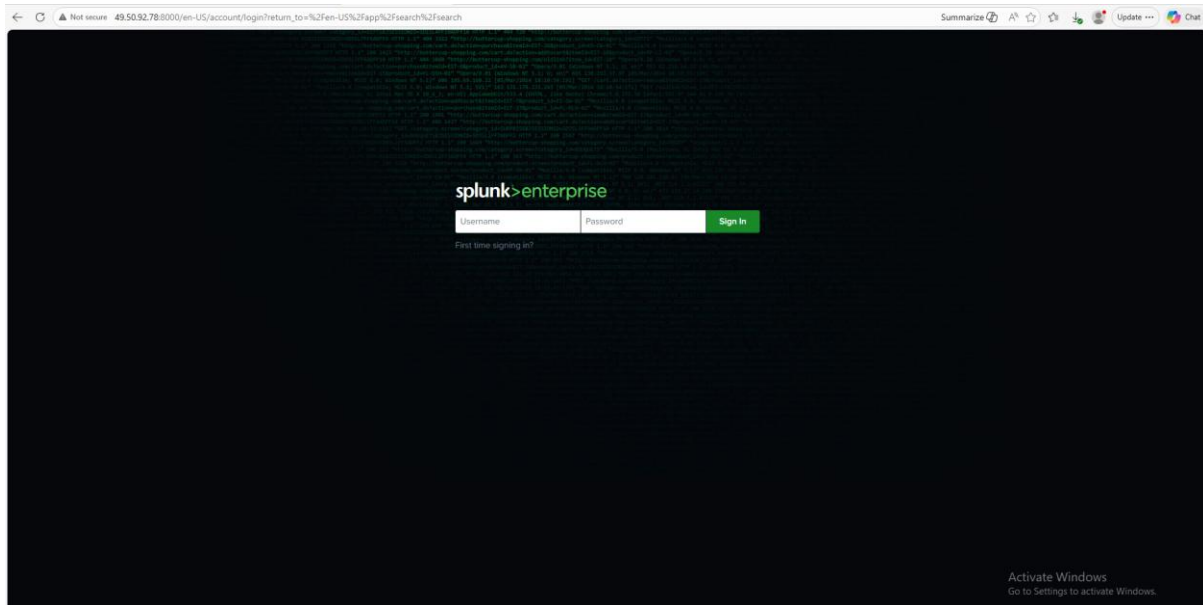
Splunk Service Verification

The Splunk service was started and verified using:

```
/opt/splunk/bin/splunk start --run-as-root
```

The service status confirmed that Splunk was running properly.

Fig: Splunk Login Page



Receiving Port Configuration

To receive logs from remote systems, Splunk was configured to listen on port 9997.

This port is used by Splunk Universal Forwarders to send log data.

Verification confirmed that the receiving port was active and listening for incoming connections.

Port 9997 Configuration

```
root@splunk-server:~# sudo ss -tulnp | grep 9997
tcp    LISTEN 0      128          0.0.0.0:9997      0.0.0.0:*        users:(("splunkd",pid=99954,fd=140))
root@splunk-server:~#
```

Phase 2 - Target Server Setup

SSH Server Deployment

A separate Ubuntu machine was deployed as the target system.

Configuration

VM Name: target-server

IP Address: 192.168.0.6

Purpose: SSH Attack Target

OpenSSH Server was installed and verified successfully.

A test user account was created for brute-force testing.

Log Source

Authentication events generated by SSH are stored in:

/var/log/auth.log

These logs are used as the primary source for attack detection.

Fig:-SSH Service Status

```
root@splunk-server:~# sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-06-05 06:19:08 UTC; 4 days ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 52627 (sshd)
      Tasks: 1 (Limit: 38356)
    Memory: 9.7M
       CPU: 5min 57.292s
   CGroup: /system.slice/ssh.service
           └─52627 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 09 09:45:26 splunk-server sshd[865794]: Failed password for root from 58.229.180.110 port 53112 ssh2
Jun 09 09:45:26 splunk-server sshd[865794]: Connection closed by authenticating user root 58.229.180.110 port 53112 [preauth]
Jun 09 09:45:27 splunk-server sshd[865796]: Failed password for root from 58.229.180.110 port 52194 ssh2
Jun 09 09:45:27 splunk-server sshd[865796]: Connection closed by authenticating user root 58.229.180.110 port 52194 [preauth]
Jun 09 09:45:28 splunk-server sshd[865799]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=125.141.231.18 user=root
Jun 09 09:45:30 splunk-server sshd[865799]: Failed password for root from 125.141.231.18 port 64504 ssh2
Jun 09 09:45:30 splunk-server sshd[865799]: Connection closed by authenticating user root 125.141.231.18 port 64504 [preauth]
Jun 09 09:45:49 splunk-server sshd[865847]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=58.229.180.107 user=root
Jun 09 09:45:51 splunk-server sshd[865847]: Failed password for root from 58.229.180.107 port 59189 ssh2
Jun 09 09:45:52 splunk-server sshd[865847]: Connection closed by authenticating user root 58.229.180.107 port 59189 [preauth]
root@splunk-server:~#
```

Target Server

```
root@target-server:~# hostname
target-server
root@target-server:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:50:56:01:30:04 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.0.6/24 brd 192.168.0.255 scope global ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::250:56ff:fe01:3004/64 scope link
        valid_lft forever preferred_lft forever
root@target-server:~#
```

Phase 3 - Log Collection

Splunk Universal Forwarder

Splunk Universal Forwarder was installed on the target server.

The forwarder was configured to send logs to:

192.168.0.5:9997

The following file was monitored:

/var/log/auth.log

After configuration, authentication events became visible in Splunk, confirming successful log ingestion.

Universal Forwarder Configuration

```
root@target-server:~# sudo /opt/splunkforwarder/bin/splunk status
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
splunkd is running (PID: 174399).
splunk helpers are running (PIDs: 174455).
root@target-server:~#
```

Logs Visible in Splunk

The screenshot displays the Splunk Enterprise search interface. The search bar contains the query `index=* "Failed password"`. The results show 15 events from 6/8/26 11:30:00.000 AM to 6/9/26 12:28:36.000 PM. The events are listed in a table with columns for Time and Event. The events show failed password attempts for various users (adminuser, testuser) from 192.168.0.7 port 45932 ssh2. The source is `/var/log/auth.log` and the sourcetype is `syslog`.

Time	Event
6/9/26 11:24:03.000 AM	Jun 9 05:54:03 target-server sshd[181335]: Failed password for invalid user adminuser from 192.168.0.7 port 45932 ssh2
6/9/26 11:23:57.000 AM	Jun 9 05:53:57 target-server sshd[181335]: Failed password for invalid user adminuser from 192.168.0.7 port 45932 ssh2
6/9/26 11:23:51.000 AM	Jun 9 05:53:51 target-server sshd[181335]: Failed password for invalid user adminuser from 192.168.0.7 port 45932 ssh2
6/9/26 11:20:53.000 AM	Jun 9 05:50:53 target-server sshd[181331]: message repeated 2 times: [Failed password for testuser from 192.168.0.7 port 45930 ssh2]
6/9/26 11:20:28.000 AM	Jun 9 05:50:28 target-server sshd[181331]: Failed password for testuser from 192.168.0.7 port 45930 ssh2
6/9/26 11:16:35.000 AM	Jun 9 05:46:35 target-server sshd[181326]: message repeated 2 times: [Failed password for testuser from 192.168.0.7 port 45928 ssh2]
6/9/26 11:16:23.000 AM	Jun 9 05:46:23 target-server sshd[181326]: Failed password for testuser from 192.168.0.7 port 45928 ssh2
6/9/26 11:14:35.000 AM	Jun 9 05:44:35 target-server sshd[181326]: message repeated 2 times: [Failed password for testuser from 192.168.0.7 port 45926 ssh2]
6/9/26 11:14:27.000 AM	Jun 9 05:44:27 target-server sshd[181326]: Failed password for testuser from 192.168.0.7 port 45926 ssh2
6/9/26 12:28:36.000 PM	Jun 8 12:02:19 target-server sshd[180256]: message repeated 2 times: [Failed password for testuser from 192.168.0.7 port 45924 ssh2]

Phase 4 - Attack Simulation

Attacker Machine

A dedicated attacker VM was deployed to simulate SSH brute-force attacks.

Configuration

VM Name: attacker-server

IP Address: 192.168.0.7

Tool Used: Hydra

Purpose: Generate Failed SSH Logins

Challenge Encountered

During testing, Hydra failed with a connection refused error.

Investigation revealed that SSH was listening on port 2232 instead of the default SSH port 22.

After updating the Hydra configuration, the attack was executed successfully.

The attack generated multiple failed authentication events.

Examples:

Failed password for testuser

Failed password for invalid user

Phase 5 - Detection Engineering

Detection Logic

To identify brute-force attempts, a simple SPL query was created.

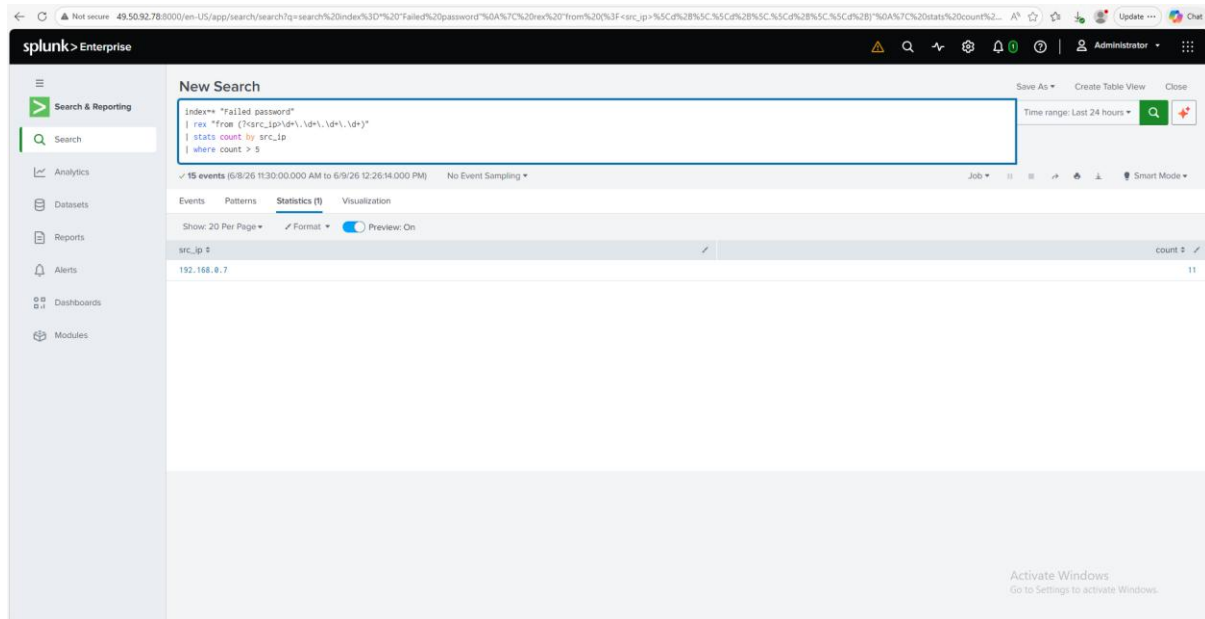
SPL Query

```
index=* "Failed password"
```

This query searches for failed SSH authentication events collected from the target server.

The detection successfully identified all attack-generated events.

SPL Search Results



The screenshot shows the Splunk Enterprise interface. The search bar contains the following SPL query: `index=* "failed password" | rex "frcm (?<src_ip>[a-z0-9]+\.[a-z0-9]+\.[a-z0-9]+\.[a-z0-9]+)" | stats count by src_ip | where count > 5`. The search results are displayed in a table with the following columns: `src_ip` and `count`. The table shows one result: `192.168.0.7` with a count of `11`. The interface also includes a sidebar with navigation options like Search & Reporting, Analytics, Datasets, Reports, Alerts, Dashboards, and Modules. The top navigation bar shows the user is logged in as Administrator.

src_ip	count
192.168.0.7	11

Phase 6 - Real-Time Alerting

Alert Configuration

A real-time alert was created in Splunk.

Alert Settings

Alert Name: SSH Brute Force Detection

Type: Real-Time

Trigger: Per Result

Whenever a failed authentication event is detected, Splunk automatically generates an alert.

Alert Configuration

Edit alert

×

Alert title

SSH Brute Force

Description

Schedule & trigger conditions

Trigger actions

*Alert type

Scheduled

Real-time

Expires after

24

hour(s)

*Trigger alert when

Per-result

Turn on throttle

☐

If throttle is on, then after an alert is triggered, subsequent alerts are suppressed for the duration of the throttle period. [Throttle alerts documentation](#)

Cancel

Save

Triggered Alert

splunk> Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

Alerts

Alerts set a condition that triggers an action, such as sending an email that contains the results of the triggering search to a list of people. Click the name to view the alert. Open the alert in Search to refine the parameters.

Q filter

This app's ▾

3 Alerts

Show: 20 per page ▾

1 ▾

of 1 pages

<

>

Title ↑	Actions	Next scheduled time ↑	Owner ↑	App ↑	Sharing ↑	Status ↑
> SSH Brute Force	Open in search Edit ▾	Jun 9, 2026 12:32:00 PM	admin	search	Private	✓ Enabled
> SSH Brute Force Detection	Open in search Edit ▾	Jun 9, 2026 12:32:00 PM	admin	search	Private	✓ Enabled
> SSH BruteForce Detection	Open in search Edit ▾	Jun 9, 2026 12:32:00 PM	admin	search	Private	✓ Enabled

Activate Windows
Go to Settings to activate Windows.

Phase 7 - Shuffle SOAR Integration

SOAR Deployment

Shuffle SOAR was deployed on a dedicated Ubuntu server.

Docker containers for:

- Backend
- Frontend
- Worker

were verified and running successfully.

Workflow Design

Webhook Trigger

↓

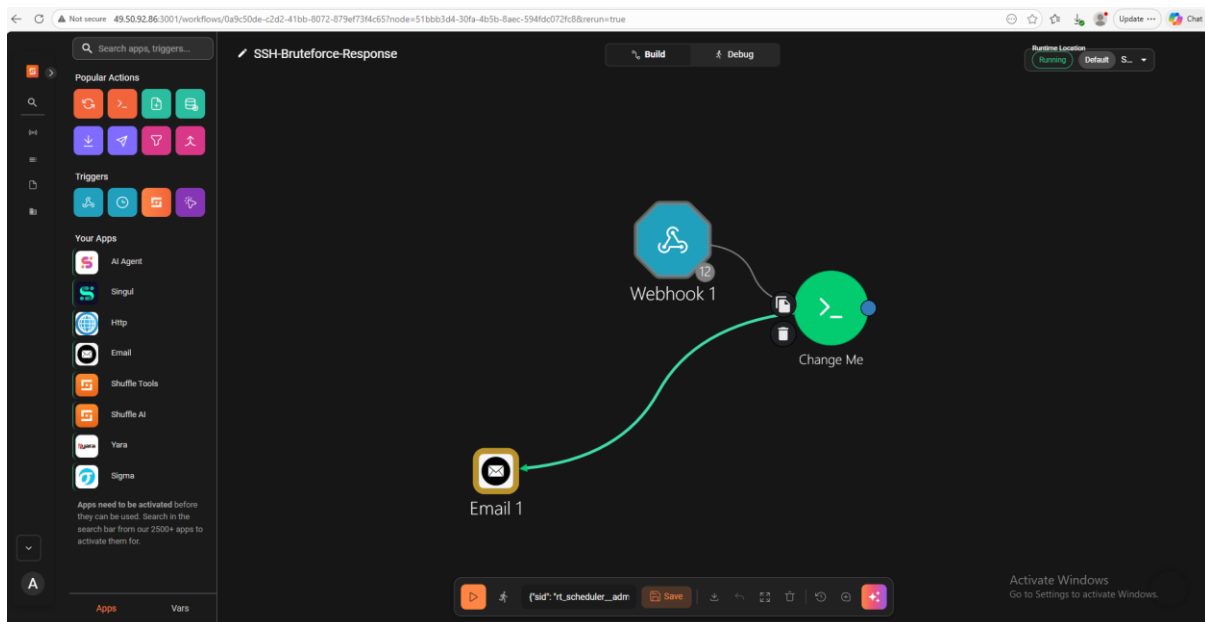
Python Action

↓

Email Notification

The workflow was designed to automatically process alerts received from Splunk.

Workflow Design



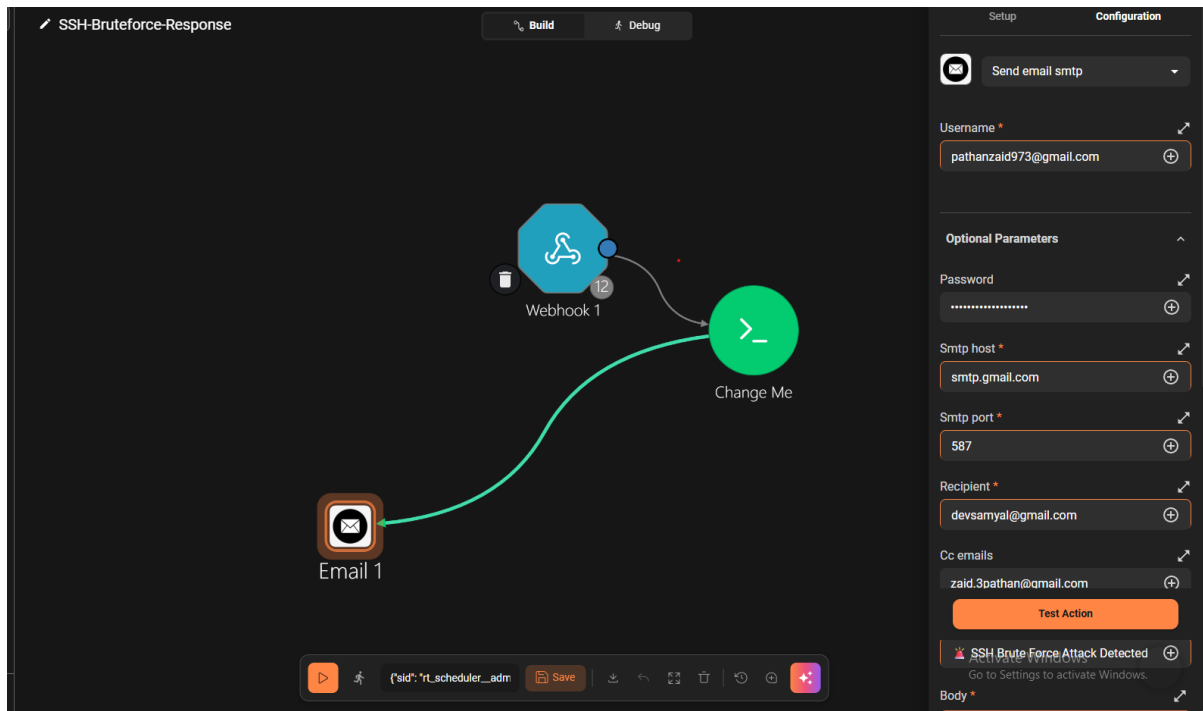
Splunk Integration

A webhook endpoint generated by Shuffle was added to the Splunk alert action.

When an alert is triggered, Splunk automatically sends event data to Shuffle.

Testing confirmed successful communication between both platforms.

Webhook Configuration



Phase 8 - Automated Email Response

SMTP Configuration

Gmail SMTP was configured inside Shuffle.

Challenge Encountered

Initial authentication attempts failed with:

530 Authentication Required

Resolution

- Enabled Gmail Two-Factor Authentication
- Generated Gmail App Password
- Updated SMTP Credentials

After applying these changes, email delivery worked successfully.

Notification Example

Subject:

@ SSH Brute Force Attack Detected

The workflow now automatically sends security notifications whenever an attack is detected.

SMTP Configuration

The screenshot shows the 'Configuration' page for the 'Send email smtp' workflow. The 'Username' field is set to 'pathanzaid973@gmail.com'. The 'Optional Parameters' section is expanded, showing 'Password' (masked), 'Smtp host' (smtp.gmail.com), 'Smtp port' (587), 'Recipient' (devsamyal@gmail.com), and 'Cc emails' (zaid.3pathan@gmail.com). A 'Test Action' button is visible. At the bottom, a preview of the email body shows a red alert icon and the text 'SSH Brute Force Attack Detected'.

Email Received

The screenshot shows an email received from 'pathanzaid973@gmail.com' to 'devsamyal, zaid.3pathan'. The email subject is 'SSH Brute Force Attack Detected'. The body contains the following sections:

- Alert Summary**
 - Alert Name: SSH Brute Force Detection
 - Severity: High
 - Source IP: 192.168.0.7
 - Target Host: target-server
 - Target IP: 192.168.0.6
 - Status: Detected & Reported
- Attack Details**

Multiple failed SSH authentication attempts were detected against the target server.

 - Repeated failed login attempts
 - Password guessing activity observed
 - Potential unauthorized access attempt
- Recommended Actions**
 - Investigate the source IP address.
 - Review SSH authentication logs.
 - Verify whether a successful login followed failed attempts.
 - Block the attacker IP if confirmed malicious.
 - Check the target server for compromise indicators.
- Automation Information**

Detection Platform: Splunk Enterprise
Response Platform: Shuffle SOAR
Workflow: SSH-Bruteforce-Response
Execution Status: Successful

The email interface includes a 'Reply' button and a 'Reply to all' button.

Results

The project successfully achieved the following:

- ✓ SSH Brute Force Attack Simulation
 - ✓ Linux Authentication Log Collection
 - ✓ Centralized Logging with Splunk
 - ✓ Real-Time Attack Detection
 - ✓ Alert Generation
 - ✓ Splunk to Shuffle Integration
 - ✓ Automated SOAR Workflow
 - ✓ Email-Based Incident Notification
 - ✓ End-to-End Detection and Response Pipeline
-

Challenges Faced

Issue	Resolution
SSH Connection Refused	Identified SSH Port 2232
SMTP Authentication Failure	Used Gmail App Password
Log Collection Issues	Verified Forwarder Configuration
Alert Testing	Validated SPL Query and Trigger Conditions

Future Improvements

- Automatic IP Blocking
- Wazuh Active Response Integration
- Slack Notifications
- Microsoft Teams Notifications
- ServiceNow Ticket Creation
- Threat Intelligence Enrichment
- Automated Endpoint Isolation
- Analyst Dashboard

Conclusion

This project demonstrates a complete SIEM and SOAR workflow capable of detecting SSH brute-force attacks and responding automatically through real-time alerting and email notifications.

The implementation provides a practical example of how security monitoring, detection engineering, and response automation can be combined to reduce incident response time and improve SOC operations.